

BOGOMOL'NYI EQUATIONS FOR NON-ABELIAN CHERN-SIMONS-HIGGS THEORIES

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Received 14 December 1990

We study the non-abelian Chern-Simons theory with spontaneous symmetry breaking. We find Bogomol'nyi type or self-dual equations for a particular choice of the Higgs potential; the corresponding vortex solutions for $G = \text{SU}(2)$ carry both quantized electric charge $Q = -en/2$ and angular momentum $J = nm^2/4$ (e is the fundamental charge, n is an integer associated with the Chern-Simons coefficient, and m is another integer which labels the topological sector). We propose a straightforward generalization to the $\text{SU}(N)$ case.

Recently, there has been considerable interest in vortex solutions to the pure Chern-Simons-Higgs theory, i.e., scalar fields minimally coupled to a vector field A_μ , which dynamics is governed by the Chern-Simons (CS) term.^{1,2}

The Chern-Simons term, peculiar to 3 space-time dimensions,

$$L_{\text{CS}} = -\frac{\mu}{2} \text{tr} \epsilon^{\alpha\beta\gamma} \left(\partial_\alpha A_\beta A_\gamma - \frac{2e}{3} A_\alpha A_\beta A_\gamma \right) \quad (1)$$

was originally added to the usual gauge field Lagrangian providing a “topological mass” for the excitations of the resulting theory.^{3,4} Later on,^{5–8} L_{CS} was considered as the entire gauge field action for models now called “Chern-Simons theories”, for which relevant applications to high- T_c superconductivity,⁹ computation of topological invariants of 3-manifolds⁶ and connections with 2-dimensional conformal field theories⁷ have been discussed.

The addition of a CS-term to abelian and non-abelian spontaneously broken gauge theories also produces interesting effects: magnetic vortices, characteristic of gauge-Higgs systems, acquire electric charge.^{10–12} This fact follows from the fundamental relation between electric charge Q and magnetic flux Φ ,

$$Q = \mu\Phi, \quad (2)$$

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which can be derived from the time-component of the gauge field equations of motion when a CS-term is present. Relation (2) means that, under the influence of the CS-term, each charge becomes a flux point and vice versa. This phenomenon is at the root of quantum statistics transmutation exploited in anyon superconductivity.⁹ Indeed, the statistical parameter $\Delta\theta$, arising when objects carrying magnetic flux and electric charge in $2 + 1$ dimensions wind one around another, is given by¹³

$$\Delta\theta = Q\Phi. \quad (3)$$

For solutions carrying both quantized charge and flux, this relation leads to fractional statistics.

As stated above, charged vortex solutions have been found in both abelian¹⁰ and non-abelian^{11,13} gauge theories with CS-term. One can then pose the question whether these solutions survive the elimination of the $F_{\mu\nu}^2$ term from the “spontaneously broken symmetry Lagrangian”, the resulting CS theory being that of relevance in anyon superconductivity.¹³ This question has been answered affirmatively in the abelian case by Jackiw and Weinberg.² Moreover, these authors, as well as Hong, Kim and Pac,¹ have shown that Bogomol’nyi type equations¹⁴ can be derived provided a specific form for the Higgs potential is assumed.

In this work we present an extension of this result to the non-abelian case, finding first order self-dual equations which can be seen to have vortex-like solutions carrying both electric and magnetic charges. It is interesting to note that in consequence of the quantization of the CS coefficient μ in the non-abelian case,³

$$\mu = \frac{e^2 n}{4\pi}, \quad (4)$$

both the electric charge Q and the angular momentum J become quantized. For example, for the SU(2) case, for which the magnetic flux of the vortex solution is just $\Phi = -2\pi m/e$, one has

$$\begin{aligned} Q &= -\frac{en}{2}, \\ J &= \frac{nm^2}{4}. \end{aligned} \quad (5)$$

With this, the statistical parameter $\Delta\theta$ is

$$\Delta\theta = \frac{\pi nm^2}{2}, \quad (6)$$

and then for the $n = 1, m = 1$ case we see that the SU(2) vortex has electric charge $Q = -e/2$ and angular momentum $J = 1/4$, thus obeying half-Fermi statistics.

Let us start by briefly discussing the conditions under which non-abelian vortices exist.^{11,12} First, the gauge symmetry must be spontaneously broken via Higgs fields leaving invariant the vacuum under a certain subgroup H of the gauge group G . Second, finite energy requirements impose certain asymptotic behavior on gauge

fields and Higgs scalars.

Concerning the first condition, for $G = \text{SU}(N)$, vortex solutions have been found for maximum symmetry breaking. This can be achieved by taking N Higgs field in the adjoint representation and choosing an adequate symmetry breaking potential. In this case $H = Z_N$ and the relevant homotopy group is $\Pi_1(G/H) = \Pi_1(\text{SU}(N)/Z_N) = Z_N$. One then has, for $\text{SU}(N)$, $N-1$ topologically inequivalent solutions (compare with the abelian case, $G = \text{U}(1)$, $H = I$ and $\Pi_1(G/H) = Z$).

Let us consider for simplicity the $\text{SU}(2)$ case. The Lagrangian describing our model reads

$$L = \frac{1}{2} D_\mu \phi D_\mu \phi + \frac{1}{2} D_\mu \psi D_\mu \psi - V[\phi, \psi] + \frac{\mu}{4} \varepsilon_{\mu\nu\alpha} [\mathbf{F}^{\mu\nu} \cdot \mathbf{A}^\alpha - \frac{e}{3} \mathbf{A}^\alpha \cdot (\mathbf{A}^\mu \wedge \mathbf{A}^\nu)], \quad (7)$$

where, as stated above, the dynamics of the gauge field $\mathbf{A}_\mu$ is given by a CS-term and no $\mathbf{F}_{\mu\nu}^2$ term is considered. Two Higgs fields ϕ and ψ are present in order to have maximum symmetry breaking. The covariant derivative for these scalar fields is

$$D_\mu \phi = \partial_\mu \phi + e \mathbf{A}_\mu \wedge \phi. \quad (8)$$

The symmetry breaking potential $V[\phi, \psi]$ has to be chosen so as to ensure complete symmetry breaking¹¹ and this can be implemented by taking it as

$$V[\phi, \psi] = V_1[\phi] + V_2[\psi] + g(\phi \cdot \psi)^2. \quad (9)$$

Here, V_1 and V_2 have to be chosen so that

$$\begin{aligned} \lim_{\rho \rightarrow \infty} \phi &= \eta, \\ \lim_{\rho \rightarrow \infty} \psi &= \eta'. \end{aligned} \quad (10)$$

The third term in the potential, Eq. (9), ensures no parallelism in isospace between ϕ and ψ so as to have a topologically stable vortex.

As it is well-known, finite energy requires the following behavior at infinity for gauge and scalar fields,

$$\begin{aligned} \lim_{\rho \rightarrow \infty} A_\mu^a \frac{\sigma_a}{2} &= \frac{1}{e} g^{-1} \partial_\mu g, \\ \lim_{\rho \rightarrow \infty} D_\mu \phi &= 0 \end{aligned} \quad (11)$$

(σ_a are Pauli matrices).

If we call φ the angle characterizing the direction at infinity, a mapping $g(\varphi)$ belonging to the m homotopy class, satisfies, when one turns around a closed

contour,

$$g_m(2\pi) = \exp\left(\frac{2\pi i m}{N}\right) g_m(0),$$

$$g_m \in \text{SU}(N). \quad (12)$$

This can be achieved just by taking g in the Cartan subgroup of the gauge group. In the present $\text{SU}(2)$ case this corresponds to

$$g = \exp i\sigma_3 \Omega(\varphi) \quad (13)$$

and

$$\Omega(2\pi) - \Omega(0) = (2k + m)\pi, \quad (14)$$

where $k \in \mathbb{Z}$ and $m = 0, 1$. More general configurations in the same topological sector can be obtained by gauge rotations.

Let us note at this point that $\text{SU}(2)$ vortices have been found^{11,12} by taking one of the Higgs fields as a constant one everywhere, i.e., playing no dynamical role and not contributing to the total action (but ensuring complete symmetry breaking). It seems reasonable to expect that any other configuration with a non-constant Ψ should lead to a greater energy. We shall then follow this point of view and look for Bogomol'nyi equations under the following conditions¹⁵ (valid everywhere in space):

$$\begin{aligned} D_\mu \Psi &= 0, \\ \Psi^2 &= \eta'^2, \\ \Phi \cdot \Psi &= 0. \end{aligned} \quad (15)$$

With this, the last two terms in Eq. (9) vanish identically.

The existence of Bogomol'nyi equations usually requires a specific form for the Higgs potential. For instance, for abelian Higgs model¹⁴ one must choose a fourth-order potential and this particular relation between coupling constants must hold: $\lambda = e^2/8$. When a CS-term is present,^{1,2} the Higgs potential must be of sixth-order and an analogous relation between coupling constants must hold. As we shall see, the same holds in the non-abelian case but, for the moment, we leave $V_1[\Phi]$ free and look for Bogomol'nyi equations in the usual way,¹⁴ i.e., starting from the energy-momentum tensor which reads

$$T_{\mu\nu} = D_\mu \Phi D_\nu \Phi - g_{\mu\nu} \left(\frac{1}{2} D_\alpha \Phi D^\alpha \Phi - V_1[\Phi] \right). \quad (16)$$

The energy functional for a static configuration is then,

$$E = \int d^2x T_{00} = \int d^2x \left[\frac{e^2}{2} (\mathbf{A}_0 \wedge \Phi)^2 + \frac{1}{2} (D_i \Phi)^2 + V_1[\Phi] \right]. \quad (17)$$

We can now eliminate $\mathbf{A}_0$ from Eq. (17) by using the constraint equation which follows from the equations of motion. Indeed, the time-component of the gauge-field equation of motion reads (taking into account conditions (15))

$$e(\phi \wedge D^0 \phi) = \frac{\mu}{2} \varepsilon^{ij} \mathbf{F}_{ij} \equiv \mu \mathbf{H}. \quad (18)$$

Then

$$\mathbf{A}_0 \wedge \phi = \frac{\mu}{e^2 \phi^2} \mathbf{H} \wedge \phi, \quad (19)$$

which immediately implies

$$\phi \cdot \mathbf{H} = 0, \quad (20)$$

and, hence, we find for the energy

$$E = \int d^2 x \left[\frac{\mu^2}{2e^2 \phi^2} H^2 + \frac{1}{2} (D_i \phi)^2 + V_1[\phi] \right]. \quad (21)$$

This expression can be written in the form

$$\begin{aligned} E = \int d^2 x & \left[\frac{1}{4} (D_i \phi \pm \varepsilon_{ij} \check{\psi} \wedge D_j \phi)^2 \right. \\ & + \frac{\mu^2}{2e^2 \phi^2} (\mathbf{H} \mp \frac{e^3}{2\mu^2} |\phi|^2 (|\phi|^2 - \eta^2) \check{\psi})^2 \\ & \left. + (V_1[\phi] - \frac{e^4}{8\mu^2} \phi^2 (\phi^2 - \eta^2)^2) \mp \frac{e}{2} \eta^2 (\check{\psi} \cdot H) \right] \end{aligned} \quad (22)$$

where conditions (15) have been repeatedly used (see Ref. 15 for details).

Now, the last term in Eq. (22) can be related to the topological charge. Using Eqs. (13) and (14) one can easily see that

$$A_\varphi^1 = A_\varphi^2 = 0; \quad A_\varphi^3 = \frac{d\Omega^3(\varphi)}{d\varphi}. \quad (23)$$

Moreover, from Eq. (15) one also has

$$\lim_{\rho \rightarrow \infty} \frac{d\check{\psi}}{d\varphi} = 0 \quad (24)$$

because

$$\lim_{\rho \rightarrow \infty} \check{\psi} \cdot \mathbf{A}_\varphi = \lim_{\rho \rightarrow \infty} \check{\psi}^3 A_\varphi^3. \quad (25)$$

With this, using Eq. (14) and choosing a gauge such that $e^{ij} \mathbf{A}_i \wedge \mathbf{A}_j = 0$ (see Ref. 15 for details), one has for the last term in Eq. (22)

$$\frac{e\eta^2}{2} \int d^2x \, \check{\psi} \cdot H = \frac{e\eta^2}{2} \check{\psi}_\infty^3 (2k+m)\pi . \quad (26)$$

Note that this quantity is related to the magnetic flux Φ as defined in Ref. 11:

$$\Phi = \int d^2x \, \mathbf{H} \cdot \check{\psi} . \quad (27)$$

We then see that the energy E is bounded by

$$E \geq e^2 \eta^2 \pi |\check{\psi}_\infty^3 (2k+m)| , \quad (28)$$

provided the potential $V_1[\phi]$ is chosen as

$$V_1[\phi] = \frac{e^4}{8\mu^2} \phi^2 (\phi^2 - \eta^2)^2 . \quad (29)$$

We shall consider $V_1[\phi]$ in this form. Then, we have found that, for this potential, a bound for the energy of the non-abelian Chern-Simons-Higgs theory defined by Lagrangian (7) is attained when the following Bogomol'nyi equations hold,

$$\begin{aligned} D_i \phi &= \mp \varepsilon_{ij} \check{\psi} \wedge D_j \phi , \\ \mathbf{H} &= \pm \frac{e^3}{2\mu^2} \phi^2 (\phi^2 - \eta^2) \check{\psi} , \\ D_i \Psi &= 0 . \end{aligned} \quad (30)$$

Concerning the choice (29) for the Higgs potential, let us note that, as it happens in the abelian case, the Bogomol'nyi limit forces not just a condition on the coupling constant ($\lambda = e^4/8\mu^2$) but also the necessity of a sixth order Higgs potential. Of course, this is in accord with the renormalizability condition in 2+1 dimensions.

As in the usual non-abelian case (i.e., with an $F_{\mu\nu}^2$ term instead of a Chern-Simons term) the bound for the energy (28) is not a topological invariant quantity,¹⁵ as it is in the abelian case. The presence of k in this equation implies the presence of different values of the energy in the Bogomol'nyi limit for different elements (different k 's) belonging to the same class (labeled by m). This is related to the fact that the topological charge of $SU(N)$ vortices is defined modulo N but, as signaled in Ref. 16, physical quantities may depend on the actual value of the magnetic flux associated with this topological charge.

For the class labeled by $m = 0$, the most stable solution to Eqs. (30) is that associated with $k = 0$, i.e., the vacuum. For the class labeled by $m = 1$, the most stable solutions are those associated with $k = -1$ and $k = 0$.

In respect to the solutions to system (30), let us note that the axially symmetric ansatz¹¹

$$\mathbf{A}_\varphi = \frac{1}{e}(a(\rho) - 1) \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \Phi = \eta g(\rho) \begin{pmatrix} \cos \varphi \\ \sin \varphi \\ 0 \end{pmatrix} \Psi = \eta' \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad (31)$$

corresponding to an $m = -1$ vortex, separates the angular dependence leading to radial equations which coincide with those found by Jackiw and Weinberg for the abelian case,²

$$g' = \pm \frac{ag}{\rho}, \quad \frac{a'}{\rho} = \pm \frac{e^4 \eta^4}{2\mu^2} g^2 (g^2 - 1). \quad (32)$$

All features of the vortex-like solutions thus correspond, for this special ansatz, to those found in Ref. 2. In particular, the peculiar form of the magnetic field for these CS vortices is concentrated in a ring surrounding the zero of the Higgs field with its maximum occurring when $\varphi^2 = \eta^2/2$.

Concerning the charge of the vortex solution, it can be easily computed using the fundamental relation, Eq. (2), once we obtained the flux Φ through Eq. (27); it results in

$$Q = -\frac{en}{2}. \quad (33)$$

The angular momentum follows from the expression (16) for the energy-momentum tensor and it becomes

$$J = \frac{nm^2}{4} \quad (34)$$

as previously stated.

Let us conclude by noting that the $SU(N)$ generalization of the $SU(2)$, Eqs. (30), can be easily obtained following the lines of Ref. 15. Furthermore, general expressions for the charge Q and the angular momentum J of the vortex solutions can be obtained and they are

$$Q = -\frac{ne}{2}, \quad J = \frac{nm^2}{2} \frac{N-1}{N}, \quad (35)$$

where n is an integer related to the CS coefficient and m is another integer labelling the topological sector. The statistical parameter results in

$$\Delta\theta = \pi nm^2 \frac{N-1}{N}, \quad (36)$$

leading to $\Delta\theta = 2\pi J$.

Acknowledgments

F. A. S. wishes to thank the LPTHE for hospitality and CNRS (France) and Fundación Antorchas (Argentina) for financial support.

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